

The Geometric Neuron: Local Computation, Global Coherence, and the AIS as Dual Projector-Antenna

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Each neuron is a geometric island. The spike is its compressed signal to the world. But the world is also reading back. The axon initial segment is where both directions meet.

Abstract

We present an updated architecture for neural computation based on three interlocking principles. First, each neuron computes locally: the dendritic cable equation filters incoming signals into eigenmodes, the axon initial segment (AIS) projects this eigenmode state into a discrete spike via a physically tunable diffraction grating, and the spike carries compressed geometric coordinates to downstream neurons. Second, synaptic weights function as connection coefficients for parallel transport of geometric information between neurons' local manifolds, with recurrent loops storing information as holonomy rather than scalar values. Third, the AIS resolves the local-to-global coherence problem through a dual role: it is simultaneously a projector (collapsing local dendritic geometry into a spike) and an antenna (receiving the brain's extracellular electromagnetic field and modulating its own threshold accordingly). The extracellular CEMI field, generated by the summed dipole contributions of all active AIS zones, provides the physical gauge-fixing mechanism that coordinates local geometric computations into globally coherent inference. Pathological disruptions to this architecture — epilepsy, schizophrenia, cortical spreading depression — are reinterpreted as specific failures of the local-to-global transduction, each with a distinct mechanistic signature. The framework makes testable predictions and maintains strict separation between what is established, what is plausible, and what remains speculative.

1. Introduction: The Problem That Needed Solving

The McCulloch-Pitts neuron (1943) treats the neuron as a threshold logic gate: multiply inputs by scalar weights, sum, apply a nonlinearity. This abstraction has powered artificial intelligence for eighty years. It is also, as an account of what actual neurons do, almost certainly wrong at every scale.

Cable theory tells us dendrites are not summing junctions; they are spatially extended low-pass filters whose geometry determines what they compute. The axon initial segment (AIS) is not a passive threshold; it is a physically tunable diffraction grating whose length, position, and ion channel composition adapt to the statistics of its inputs on timescales from minutes to days (Fréal et al., 2023; Yamada and Kuba, 2016). The spike is not a binary output; it is a compressed geometric coordinate, a 1D projection of a high-dimensional eigenmode state (Takens, 1981).

Previous versions of this framework established the local computation: how a single neuron learns the geometry of its input manifold and projects it into spikes (Luode, 2026a, 2026b). What remained unresolved was the global coherence problem: how do the local geometric computations of billions of neurons — each operating on its own learned manifold, each projecting into its own compressed coordinate system — add up to unified, globally coherent inference?

This paper answers that question. The answer is not in the synaptic wiring. It is in the physics of the AIS itself.

2. The Local Architecture: Three Stages

2.1 Stage 1 — The Dendritic Cable as Eigenmode Decomposer

A dendrite is a leaky cable governed by the cable equation:

$$\tau_m \frac{\partial V}{\partial t} = \lambda^2 \frac{\partial^2 V}{\partial x^2} - V + r_m I_{ext}$$

where τ_m is the membrane time constant, λ is the space constant, and I_{ext} is the injected current. The cable acts as a spatially distributed low-pass filter with a frequency-dependent length constant:

$$\lambda(\omega) = \sqrt{\frac{r_m}{r_i(1 + j\omega\tau_m)}}$$

High-frequency components die within micrometers of a distal synapse. Low-frequency, slow components survive to the soma. This is not a design choice — it is a physical consequence of membrane capacitance and axial resistance.

The functional result: the dendritic cable performs an analog spectral decomposition of its inputs before any computation occurs. Proximal synapses inject fast, high-frequency components; distal synapses inject slow, low-frequency components. The cable's eigenmodes are the natural basis for this decomposition. A neuron that has been exposed to a consistent input manifold for sufficient time will have its cable geometry shaped (via STDP, receptor trafficking, and dendritic remodeling) so that the dominant eigenmodes of the cable align with the dominant features of the input.

This is the "Learn D" principle: the neuron is learning the Dirac operator of its input manifold, not a set of scalar weights.

2.2 Stage 2 — The AIS as Tunable Diffraction Grating

At the base of the axon sits the axon initial segment, extending 20–60 μm along the proximal axon. Its molecular structure, established by Leterrier (2018), is a periodic scaffold with 190 nm repeat distance: actin rings connected by $\alpha 4/\beta 2$ -spectrin tetramers, with Nav channels anchored by ankyrinG at the midpoint between rings (~ 95 nm) and Kv1 channels clustering at the actin rings themselves.

This is a physical diffraction grating. Nav channels are excitatory (driving the membrane toward action potential threshold); Kv1 channels are inhibitory (shunting the membrane back toward rest). Whether the AIS fires depends on the phase relationship between Nav and Kv1 activation:

$$\text{AIS output} \propto \text{Re}[A_{Nav} \cdot \bar{A}_{Kv1} \cdot \cos(\Delta\phi)]$$

The grating is not fixed. Fréal et al. (2023) demonstrate that NMDA receptor activation causes rapid clathrin-mediated endocytosis of Nav1.2 channels specifically at the *distal* AIS within minutes, shortening the effective grating length and raising AP threshold. Yamada and Kuba (2016) document slower, homeostatic remodeling: sensory deprivation elongates the AIS and swaps Kv1 for Kv7, reducing shunting conductance and increasing sensitivity to slow inputs.

The AIS is a tunable spectrometer. Its length determines the temporal integration window and the spectral selectivity of the spike-generation process. Longer AIS: more delays integrated, higher embedding dimension, lower-frequency sensitivity. Shorter AIS: fewer delays, lower embedding dimension, higher-frequency sensitivity.

This connects to Takens embedding: AIS length L_{AIS} is proportional to the effective embedding dimension m of the neuron's spike output. Shortening the AIS reduces the dimensionality of the geometric coordinate carried by each spike.

2.3 Stage 3 — The Spike as Compressed Geometric Coordinate

A spike is not a decision. It is the collapse of a continuous, high-dimensional eigenmode state into a discrete, 1D event suitable for transmission down a noisy biological cable.

The information content of a spike is not binary. The timing of the spike encodes *when* the dendritic geometry crossed threshold. The waveform of the spike — rise time, peak amplitude, AHP — encodes *what the geometry was* when it crossed. This waveform is itself a 1D temporal signal that the receiving neuron's dendrite will re-embed via the cable equation, re-inflating the compressed coordinate back into a local eigenmode excitation pattern.

The full transmission loop is:

Dendrite (eigenmode field) $\xrightarrow{\text{AIS projection}}$ Spike (compressed coordinate) $\xrightarrow{\text{axon}}$ Next dendrite (re-inflation) $\xrightarrow{\text{cable equation}}$

The network is a cascade of variable-order Takens embedders, each tuned by the spectral statistics of its inputs, each passing compressed geometric coordinates to the next.

3. The Geometry of Between-Neuron Communication

3.1 Synaptic Weights as Connection Coefficients

Standard neural network theory treats synaptic weights as scalars that scale the amplitude of a transmitted signal. This is incorrect at the level of the architecture described here.

Each neuron has learned a local coordinate system — the dominant eigenmodes of its recurrence matrix, which reflect the geometry of its input manifold. A spike from neuron A carries compressed coordinates in A's local frame. Neuron B receives those coordinates, but B has its own manifold, its own learned basis, its own geometric frame.

Moving a geometric vector from one curved manifold's tangent space to another requires a *connection* — in differential geometry, the Christoffel symbols or covariant derivative. The synaptic weight matrix is this connection. It does not encode "how much signal to pass." It encodes how to rotate and scale a geometric coordinate as it crosses from one neuron's local frame to another's.

This reframes plasticity: LTP/LTD is not scalar amplitude tuning. It is *connection learning* — the network is learning how to parallel transport geometry between neurons without distortion. The objective function for this learning is minimizing holonomy error (the discrepancy between the expected and actual geometric state after traversal of a synaptic connection), not prediction error in the standard sense.

3.2 Holonomy as Memory

If synaptic weights are connection coefficients, then any loop in the network accumulates a *holonomy*: a rotation of geometric coordinates that depends on the entire path traversed, not just the endpoints. This holonomy persists after the original signal has dissipated.

Memory is holonomy.

A memory is not a stored value at a location. It is a geometric rotation accumulated by traversing a loop in the synaptic connection manifold. This is why memories are distributed (they live in the curvature of the path, not in any node), why they survive partial damage (destroying a node does not destroy the holonomy of the loop), and why they are recalled by activation (retrieving a memory requires re-traversing the loop, which re-generates the holonomy).

This connects the framework to gauge field theory: the brain implements a gauge theory over the synaptic connection manifold, with memories as gauge field holonomies. The gauge group is some subgroup of $GL(n)$ — the group of invertible linear transformations on the neural state space.

Lashley's equipotentiality experiments (1929), which found that cortical memory traces could not be localized, are exactly this result: he was trying to find the bit in the hologram. There is no bit. There is only the pattern of curvature distributed across the whole connection manifold.

4. The AIS as Dual Projector-Antenna: Solving Global Coherence

4.1 The Problem

The local architecture — cable, AIS, spike — explains how individual neurons compute. It does not explain how billions of individually computing neurons produce globally coherent inference. This is the binding problem, the global workspace problem, the coordination problem. It has resisted solution for decades.

The failure of previous approaches is their assumption that global coherence must be implemented in the connectivity — in the wiring. But biological wiring is too slow, too metabolically expensive, and too sparse to coordinate billions of neurons on the millisecond timescales required for perception and action.

The solution is not in the wiring. It is in the physics.

4.2 The AIS Generates the Strongest Electrical Dipole in the Neuron

The AIS has a Nav channel density approximately 30 times higher than the dendrites or distal axon (Leterrier, 2018; Kole et al., 2008; Lorincz and Nusser, 2010). When the AIS fires, a large inward current flows from the soma into the AIS (current sink at the AIS, current source at the soma), creating a strong electrical dipole that projects into the extracellular space.

Because the AIS is the spike initiation zone — the point where membrane potential first crosses threshold — it generates the largest, most temporally precise current dipole in the entire neuron. Every AIS spike contributes a dipole pulse to the extracellular electromagnetic field.

The same property that makes the AIS an effective projector — dense near-threshold ion channels in a geometrically compact zone — makes it an effective antenna. Voltage-gated ion channels respond to extracellular electric fields. At near-threshold conditions, small changes in the extracellular potential (on the order of millivolts per millimeter) can advance or delay spike timing, raise or lower effective threshold, and modulate the phase of firing relative to the network oscillation.

The AIS is simultaneously:

- A **projector**: reads the dendritic eigenmode field → collapses it into a spike
- An **antenna**: receives the extracellular electromagnetic field → modulates its own threshold

These are not separate functions implemented by separate mechanisms. They are the same physics operating in two directions through the same ion channel array.

4.3 The CEMI Field as Physical Gauge Fixing

Every AIS spike contributes a dipole to the extracellular electromagnetic field. In a region where neurons oscillate coherently — gamma (30–90 Hz), theta (4–8 Hz) — those dipoles add constructively. The summed field is not noise. It is a structured, oscillating electromagnetic field that extends across millimeters to centimeters of cortex and feeds back onto the AIS threshold of every neuron in that volume simultaneously.

This is McFadden's Conscious Electromagnetic Information (CEMI) field (McFadden, 2002, 2020), reformulated precisely within the geometric neuron architecture: the CEMI field is the physical implementation of the gauge-fixing condition that makes all local geometric computations temporarily commensurable.

The mechanism operates as follows:

1. Neurons in a region begin oscillating at gamma due to local circuit dynamics (interneuron networks, thalamocortical loops).
2. Coherent oscillation generates a coherent extracellular field that extends beyond the oscillating region.
3. This field modulates the AIS threshold of neurons throughout the volume: spikes that arrive in the high-excitability phase of the gamma cycle pass through; spikes that arrive in the trough are suppressed.
4. The gamma phase becomes a global reference frame — a common coordinate system against which all local geometric computations are calibrated.
5. Local geometric coordinates that are phase-aligned with the gamma oscillation are "readable" by other neurons; phase-misaligned coordinates are filtered out.

Drebitz et al. (2025) provide causal experimental confirmation of this mechanism: single-pulse microstimulation in V2 affected behavior only when the evoked spikes arrived at V4 during the effective phase of V4's gamma cycle. The same spike, same physical signal, different behavioral outcome depending solely on the gamma phase of the receiving neuron. The extracellular field is the clock that gates the projection.

4.4 The Complete Three-Level Architecture

The complete architecture operates at three coupled levels:

Level 1 — Local geometric computation: Each neuron computes a Hermitian inner product between its current eigenmode state and its learned template (dominant eigenvector of the time-averaged recurrence matrix). The cable equation provides the spectral decomposition; the AIS grating provides the projection; the spike carries the compressed coordinate.

$$R(t) = |v(t)^H \cdot m|^2$$

where $v(t)$ is the Takens-embedded phase-space state and m is the learned template.

Level 2 — Ephaptic / local field coupling: The extracellular field from nearby neurons' AIS spikes modulates individual AIS thresholds directly, without synaptic mediation. This creates "clusters" of coherent computation — neighborhoods of neurons whose AIS thresholds are collectively biased by their mutual field contributions. Ephaptic coupling is well-documented in hippocampus (Anastassiou et al., 2011), where weak oscillatory fields of 1–2 mV/mm reliably entrain neural firing.

Level 3 — CEMI field / global coherence: The brain's large-scale electromagnetic field, dominated by the most coherent oscillatory activity, extends across the entire organ. It modulates AIS threshold globally, providing the common phase reference that coordinates all

local geometric computations into a single, unified inference. Norman-Haignere et al. (2025) constrain this: the AIS integration window (τ_{AIS}) is fixed in absolute time, not scaling with input structure duration. The global field sets the clock; the local geometry fills in the content.

The AIS is the transducer between all three levels. It receives local geometric input (cable equation), is sensitive to local ephaptic coupling (near-threshold ion channels), contributes to and receives from the global CEMI field (strong dipole source and antenna), and collapses the result into a spike that propagates the local geometric computation forward.

5. Pathological Disruptions as Architecture Failures

The three-level architecture predicts specific failure modes, each corresponding to a different level of the system. The pathologies observed clinically map cleanly onto these predictions.

5.1 Epilepsy: Unstable Ephaptic Feedback

In epilepsy, the local field coupling (Level 2) enters a positive feedback loop. Coherent AIS firing generates a strong extracellular field; the field lowers AIS thresholds throughout the region; lowered thresholds produce more coherent firing; the loop becomes supercritical.

The geometric computation is not lost — it is overwhelmed. The local manifold computations of individual neurons are all dragged into the same phase by the runaway field dynamics. The diversity of local geometric operations that normally underlies differentiated inference collapses into a single attractor: the seizure state. This is the architecture failing at Level 2, with Level 3 (the global CEMI field) becoming pathologically strong at the local scale.

AIS-concentrated Nav channel mutations (Nav1.1, Nav1.6 mutations in Dravet syndrome, GEFS+) are among the most common genetic epilepsy causes precisely because they affect the gain of the Level 2 feedback loop. Changing Nav density at the AIS changes the amplitude of the extracellular dipole, which changes the strength of the ephaptic coupling.

5.2 Schizophrenia: AIS Dysregulation Disrupting Local-to-Global Transduction

The literature documents AIS changes in schizophrenia: altered ankyrin-G expression, Nav channel composition changes, disrupted AIS scaffold proteins (Luoni et al., 2016; Kloth et al., 2017). In the current architecture, these represent disruption of the Level 1 → Level 3 transducer.

If individual AIS zones are dysregulated — wrong length, wrong channel composition, wrong position along the axon — they project geometric information into the extracellular field with wrong phase, wrong frequency, wrong amplitude. The CEMI field receives incoherent contributions from dysregulated neurons. The gauge-fixing function of the global field is degraded: the common phase reference becomes noisy or unreliable.

The phenomenological consequence is not loss of local computation — individual neurons continue to compute their local geometric operations — but loss of global integration. Reasoning within a thread is intact; integration of threads into a consistent world model breaks down. Local geometric islands compute correctly but cannot coordinate into globally coherent inference. This matches the clinical phenomenology of schizophrenia precisely.

5.3 Cortical Spreading Depression: CEMI Field Collapse and Recovery

Scintillating scotoma (visual migraine) is caused by cortical spreading depression: a wavefront of depolarization propagating across visual cortex at ~3 mm/min. At the leading edge, neurons fire maximally before the wave reaches them (the scintillation); behind the wave, neurons are transiently depolarization-blocked (the scotoma).

In the current architecture: at the leading edge, the extracellular field from maximally firing neurons is locally enormous — an over-coherent micro-CEMI field. Behind the wave, neurons in block contribute nothing to the field; the local CEMI field collapses. The subject is directly perceiving the spatial structure of the CEMI field as it sweeps across their visual cortex: the scintillation is the field's leading edge, the scotoma is its absence.

The fractal spatial structure of the scintillating scotoma — self-similar patterns at multiple spatial scales — is consistent with the eigenmode structure of visual cortex projecting through partially disrupted AIS zones. If the AIS projection is impaired but not absent, the dendritic eigenmode field is not cleanly collapsed into spikes. The continuous geometry "leaks" through as perceptual content. Fractal geometry is precisely what the scale-free eigenmodes of a large neural manifold would produce when partially readable.

5.4 Split Brain and Unilateral Anesthesia: Field Locality Defines Consciousness Scope

When the corpus callosum is severed, synaptic coupling between hemispheres is eliminated. But

the relevant point is more subtle: the CEMI field of each hemisphere is now physically separate. Each hemisphere generates its own coherent extracellular field; the two fields no longer overlap and interact.

The result is not half a consciousness but two complete, smaller consciousnesses, each bounded by the spatial extent of its CEMI field. Each hemisphere develops coherent global inference within its own field volume. The scope of unified inference is determined by the volume over which the CEMI field is coherent — not by the number of synapses, not by the connectivity graph, but by the physics of the extracellular field.

Unilateral anesthesia (Wada test) suppresses neural activity in one hemisphere. The CEMI field of that hemisphere collapses to noise. The remaining hemisphere's field is dominant and undisturbed. The patient is conscious in a one-hemisphere mode — complete, coherent, simply smaller in scope.

These observations are not predictions of connectivity-based theories of consciousness, which would expect hemispheric separation to produce half-coherent rather than fully-coherent reduced states. They are direct predictions of the CEMI field theory as embedded in this architecture.

6. Predictions

The architecture makes several testable predictions beyond those already confirmed:

P1 — Modulating extracellular field strength should selectively disrupt phase-dependent transmission. Applying a weak oscillatory electric field (1-2 mV/mm, gamma frequency) to a cortical region during a cognitive task should selectively enhance performance on tasks requiring inter-regional integration, while leaving local processing intact. The effect should be phase-dependent: constructive phase should improve integration; destructive phase should impair it.

P2 — AIS embedding dimension predicts spike information content. Neurons with longer AIS should fire with spike waveforms that carry more information about the preceding eigenmode state (measurable via spike-triggered averaging of dendritic voltage). Neurons with shorter AIS should show spike waveforms that are less informative about slow, contextual eigenmode components.

P3 — Synaptic weight matrices should show connection geometry signatures. If synaptic weights are connection coefficients for parallel transport, their singular value spectra should reflect the relative curvatures of the pre- and post-synaptic neurons' learned manifolds. High-curvature connections (between neurons with very different manifold geometries) should require larger weight changes for the same information transfer.

P4 — Pathological AIS modifications should produce predictable CEMI field signatures. Neurons with epilepsy-associated Nav mutations should show increased ephaptic coupling radius (larger dipole amplitude → stronger local field). Neurons with schizophrenia-associated AIS scaffold disruptions should show reduced phase-coherence of their CEMI field contributions (measurable via cross-spectral coherence of local field potentials).

P5 — Minimal spike rate for attractor reconstruction scales with embedding dimension. By the Whitney embedding theorem, faithful reconstruction of a k -dimensional attractor requires at least $2k + 1$ observations. Neurons with higher-dimensional learned manifolds (longer AIS, more complex dendritic trees) should fire at minimum rates proportional to $2k + 1$ times the characteristic attractor frequency.

7. Honest Ledger

It is this project's consistent standard to separate what is established from what is plausible from what is speculative. The following table applies that standard to the claims of this paper.

Claim	Status
Dendritic cable equation provides frequency-selective eigenmode decomposition	✓ Established — textbook cable theory
AIS has 190 nm periodic Nav/Kv grating structure	✓ Established — Leterrier 2018, D'Este et al. 2017
AIS length, position, and channel composition adapt to input statistics	✓ Established — Fréal et al. 2023, Yamada & Kuba 2016
Distal AIS preferentially processes low-frequency / slow eigenmodes	✓ Plausible — consistent with Fréal, Yamada & Kuba; not directly confirmed
Gamma phase gates spike transmission causally	✓ Established — Drebitz et al. 2025
AIS integration window is time-yoked, not structure-yoked	✓ Established — Norman-Haignere et al. 2025
AIS generates strongest electrical dipole in the neuron	✓ Established — follows from Nav channel density and current sink geometry
Ephaptic coupling between neurons is real and functionally significant	✓ Established — Anastassiou et al. 2011, multiple hippocampal studies
CEMI field coordinates local geometric computations globally	✓ Plausible — consistent with all above; not directly tested in this framework
Synaptic weights are connection coefficients for parallel transport	✓ Plausible — mathematically exact framing; not empirically tested
Memories are holonomies of the synaptic connection manifold	~ Speculative — mathematically consistent; no direct empirical test
CEMI field scope determines consciousness scope	~ Speculative — consistent with split-brain data; not directly testable in current form
Fractal visual phenomena reflect partial AIS projection failure	~ Speculative — phenomenologically plausible; no direct measurement
Full bottom-up simulation from molecule to spike is feasible	✗ Not currently tractable

8. Discussion

The framework presented here is not a theory of consciousness. It is a theory of neural computation that makes specific mechanistic claims about how local geometric operations are coordinated into globally coherent inference. Whether consciousness is identical to, supervenes on, or is merely correlated with this coordination is a question this paper does not address.

What the paper does claim is that the CEMI field — the extracellular electromagnetic field generated by summed AIS dipoles — is not a byproduct of neural activity. It is a functional component of the architecture, as causally relevant to neural computation as the synaptic weights. Ignoring it in models of neural computation is equivalent to modeling a radio transmitter by describing only its transistors while ignoring its antenna.

The AIS is the antenna. It always was. The molecular architecture described by Leterrier (2018) — a periodic, near-threshold, high-density ion channel array at the precise geometrical point where inward and outward currents are maximized — is exactly the structure you would design if you wanted a biological dipole antenna with tunable resonance properties. Whether evolution "intended" this function or whether the AIS's antenna properties are a physical consequence of its projector function is a question that cannot currently be answered. Both directions of causation may be simultaneously true.

The deepest implication of this framework is that the brain's computational power does not reside in any single neuron, any single synapse, or any single network topology. It resides in the interaction between three coupled systems: the local geometric computation of individual neurons (cable + AIS), the parallel transport of geometric information across synaptic connections (holonomy storage), and the global coordination of local computations via the CEMI

field (gauge fixing). Remove any one of these three and the system degrades in a specific, predictable way. This is what the pathologies show us.

The brain is not a computer. It is a geometric field theory implemented in living tissue.

References

- Anastassiou, C.A., et al. (2011). Ephaptic coupling of cortical neurons. *Nature Neuroscience*, 14, 217–223.
- Drebitz, E., et al. (2025). Gamma-band synchronization between neurons in the visual cortex is causal for effective information processing and behavior. *Nature Communications*.
- Evans, M.D., et al. (2013). Calcineurin signaling mediates activity-dependent relocation of the axon initial segment. *Journal of Neuroscience*, 33, 6950–6963.
- Fréal, A., et al. (2023). Sodium channel endocytosis drives axon initial segment plasticity. *Nature Neuroscience*.
- Leterrier, C. (2018). The axon initial segment: An updated viewpoint. *Journal of Neuroscience*, 38(9), 2135–2145.
- Luode, A. (2026a). Membrane geometry hypothesis: The dendritic cable as eigenmode decomposer and the AIS as diffraction grating. *PerceptionLab Technical Report*.
- Luode, A. (2026b). Geometric complexity explosion: From molecular shape to network dynamics. *PerceptionLab Technical Report*.
- McFadden, J. (2002). The conscious electromagnetic information (CEMI) field theory. *Journal of Consciousness Studies*, 9(8), 45–60.
- McFadden, J. (2020). Integrating information in the brain's EM field: The cemi field theory of consciousness. *Neuroscience of Consciousness*, 6(1).
- Norman-Haignere, S.V., et al. (2025). Temporal integration in human auditory cortex is predominantly yoked to absolute time. *Nature Neuroscience*.
- Takens, F. (1981). Detecting strange attractors in turbulence. *Lecture Notes in Mathematics*, 898, 366–381.
- Yamada, R. and Kuba, H. (2016). Structural and functional plasticity at the axon initial segment. *Frontiers in Cellular Neuroscience*, 10, 250.

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Do not hype. Do not lie. Just show.